

BNBU
DS 4023: Machine Learning
Spring 2026 Course Project

Description:

This is a **GROUP** project, which aims at applying machine learning algorithms (including but not limited to those covered in our lectures) to solve real-world tasks or conducting machine learning research.

The group should have 2 or 3 students, you may team up by yourselves and cross-class(session) teams are **NOT** allowed. (You need to team up and register in the ispace project group session on or before **Apr. 12, 2026**)

Project Topics:

Each GROUP will need to select a topic. There are three mainstreams of topics:

- **Application-driven projects:** These projects are driven by the need to solve a specific problem or task. Pick an application that interests you, and explore how best to **apply learning algorithms and techniques** to solve it.
- **Data-driven projects:** These projects are driven by the availability of data. You can refer to online resources, for example Kaggle or [iDataScience platform](#), for various datasets from different domains. The focus is on exploring and understanding the data, selecting appropriate features, and **apply learning algorithms** to accurately predict or classify new data points and solve real problems.
- **Research-driven projects:** These projects are driven by the desire to push the boundaries of machine learning by creating novel algorithms, enhancing current ones, or investigating uncharted territories within the field.

You are suggested to pick something that you can get excited and passionate about, e.g., either an application area that you're interested in, or pick some subfield of machine learning that you want to explore more. Alternatively, if you're already working on a research or project that machine learning might apply to, then you may already have a great project idea.

Note: You may decide to work on Deep learning project, and since our course mainly discusses **basic concepts** for machine learning, please make sure that you use **materials you learned in the class as well if you choose Deep learning project**.

- For example, you might set up logistic regression and SVM baselines, or do some data analysis using the unsupervised methods covered in class.
- Also, training deep learning models can be very time consuming, so make sure you have the necessary computing resources.
- **Deep learning model would not be considered as more superior than other machine learning models if you just repeat a model that is designed by others.**
- **Plagiarism is not tolerated, any forms of copy of existing code or report (e.g., from online platform or previous works) will be give zero mark.**
- **You need to be able to clearly state the new contribution you make on top of the existing models, if you just repeat an existing model to do a task, even though the result might**

be good, you will not get good scores for that.

To undertake the project, the following steps are essential:

1. Select one topic.
2. Survey on existing research on relevant topics by searching related keywords on an academic search engine such as: <http://scholar.google.com>.
3. Collect, read, and analyze relevant materials /data.
 - An important aspect of designing your project is to identify one or several datasets suitable for your topic of interest. Get the **benchmark datasets** and validate your learning algorithms on the benchmark datasets is preferred. We don't want you to spend much time collecting raw data.
 - If you choose to use preprepared datasets (e.g. from Kaggle or iDataScience <http://www.idatascience.cn/dataset>), we encourage you to do some data exploration and analysis to get familiar with the problem.
4. Design and implement learning algorithms and validate the proposed algorithms on benchmark/collected dataset.
 - We expect a solid methodology, comprehensive validation and detail discussion of the experimental results.
 - Replicating the results in a paper can be a good way to learn. However, instead of just replicating a paper, also try using the technique on another application, or do some analysis of how each component of the model contributes to final performance.
5. Produce a report and give the presentation.
 - A very good project report will be a publishable or nearly-publishable piece of written work. You may read some recent papers and follow the writing styles. **Refer to the sample reports for reference.**

Submission Requirement:

Upon completion, each student must submit the following materials:

1. **Project report**, your report should contain but not limited to the following content:
 - a) Motivation and background of the topic
 - b) Related works and existing techniques of the topic
 - c) Methodology
 - d) Experimental study and result analysis
 - e) Future work and conclusion
2. Link and description to the Dataset and the **implementation code**.

Deadline:

The presentation will be arranged on **Week 14**. **Each group will have 8 minutes for presentation and 4 minutes for QA.**

Report submission should be on or before **24 May, 2026**.

Assessment:

In general, projects will be evaluated based on:

- Significance. (Did the authors choose an interesting or a "real" problem to work on, or

only a small "toy" problem? Is this work likely to be useful and/or have impact?)

- The technical quality of the work. (i.e., Does the technical material make sense? Are the things tried reasonable? Are the proposed algorithms or applications clever and interesting? Do the student convey novel insight about the problem and/or algorithms?)
- The novelty of the work. (Is this project applying a common technique to a well-studied problem, or is the problem or method relatively unexplored?)
- The overall grade will be based on the performance of oral presentation(50%), report(30%) as well as codes(20%).
- The Rubrics of each part can be find in iSpace.